

Celestial Weight Calculator

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Whether you are a science fiction fan, a space enthusiast or one of the millions who have watched astronauts gamboling about the moon's surface, you may have wondered how much you would weigh on other planets in the solar system. The weight depends on the gravitational forces acting on the object, which vary from planet to planet. The Newton's law of univer-

sal gravitation says that everything that has mass attracts every other thing that has mass, pulling with a force which is directly proportional to the product of the two masses of the objects and inversely propor-

tional to the square of the distance separating their centres. The mass and radius of different celestial bodies is mentioned in Table I.

The designed embedded system takes a weight on earth and shows the weights on other celestial bodies of our solar system. The device is designed and constructed using microcontroller AT89S52. It can give better understanding about gravity. The system calculates the mass and acceleration due to gravity (g) for other celestial bodies and displays the corresponding weight (force). Fig. 1 shows the author's prototype.

Suppose your weight on earth is 50.5 kg, your weight on other celestial bodies of our solar system would be as given in Table II.

Circuit and working

Fig. 2 shows circuit diagram of the celestial weight calculator. The circuit is built around microcontroller AT89S52 (IC1), 7-segment displays LTS542 (DIS1 through DIS8), and a few transistors and switches.

Microcontroller. AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller. It provides 8k bytes of Flash, 256 bytes of RAM, 32 I/O lines, three 16-bit timer/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, on-chip oscillator and clock circuitry. Power-on reset is provided by the combination of resistor R17 and capacitor C3. Switch S6 is used for manual reset. A 12MHz crystal along with two 33pF capacitors provides basic clock frequency to the microcontroller.

Port 0 and port 2 of IC1 are used for multiplexing the 7-segment displays

TABLE I
Mass and Radius of Different Celestial Bodies

Celestial body	Mass (kg)	Radius (m)
Mercury	3.30×10^{23}	2,440,000
Venus	4.87×10^{24}	6,051,000
Earth	5.97×10^{24}	6,378,000
Moon	7.35×10^{22}	1,738,000
Mars	6.42×10^{23}	3,397,000
Jupiter	1.90×10^{27}	71,492,000
Saturn	5.69×10^{26}	60,268,000
Uranus	8.66×10^{25}	25,559,000
Neptune	1.03×10^{26}	24,764,000
Pluto	1.31×10^{22}	1,160,000
Sun	1.99×10^{30}	695,000,000

TABLE II
Weight of a 50.5kg Substance on Different Celestial Bodies

Celestial body	Weight (kg)
Sun	1367.1359
Mercury	19.088945
Venus	45.8035
Earth	50.5
Moon	8.3829994
Mars	19.038499
Jupiter	119.382
Saturn	54.54
Uranus	44.8945
Neptune	56.8125
Pluto	3.3835



Fig. 1: Author's prototype

PARTS LIST

Semiconductors:	
IC1	- AT89S52 microcontroller
IC2	- 7805, 5V regulator
BR1	- Bridge rectifier
DIS1-DIS8	- LTS542 CA 7-segment display
T1-T8	- BC557 pnp transistor
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1-R8, R17	- 10-kilo-ohm
RN1	- 10-kilo-ohm resistor network
R9-R16	- 100-ohm
Capacitors:	
C1	- 470 μ F, 35V electrolytic
C2	- 0.1 μ F ceramic
C3	- 10 μ F, 16V electrolytic
C4, C5	- 33pF ceramic
Miscellaneous:	
X1	- 230V primary to 15V, 500mA secondary transformer
X _{crystal} 1	- 12MHz crystal oscillator
S1-S24	- Tactile switch

TABLE III
Port Pin Connections of Keyboard Matrix

Port pins	P3.7	P3.6	P3.5	P3.4	P3.3	P3.2
P1.7	7	8	9	SUN	PLUTO	JUPITER
P1.6	4	5	6	VENUS	SATURN	MERCURY
P1.5	1	2	3	MOON	URANUS	NEPTUNE
P1.4	0	00	•(DP)	MARS	EARTH	

•(DP): Decimal Point